

Name:

Entry number:

(Q1)

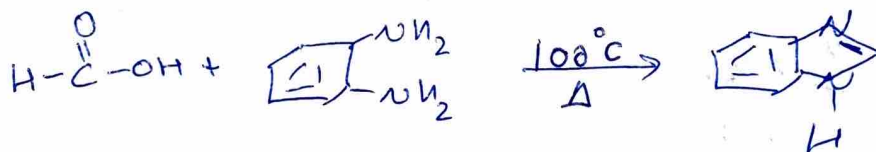
- (i) Name a benzimidazole based drug? (1)

OMEPRazole, BENIDOLYL

- (ii) Write the chemical structure of Benzimidazole. (1)



- (iii) Write the reaction used in the experiment to make Benzimidazole? (1)



- (iv) Why melting point of product is obtained after the synthesis of product? (1)

To check the purity of the compound. (If any unreacted reactant is present it will not give a sharp melting point).

- (v) How many water molecules are lost in the reaction between one molecule of
- ~~benzimidazole~~
- and one molecule of ortho-phenylene diamine? (1)

HCOOH

2

(Q2)

(vi) Which one of the following is not a condensation polymer? (1)

(A) Dacron (B) Neoprene (C) Melamine (D) Glyptal

(vii) Match the polymers given in column I with the type of linkage present in column II. (2)

Column I

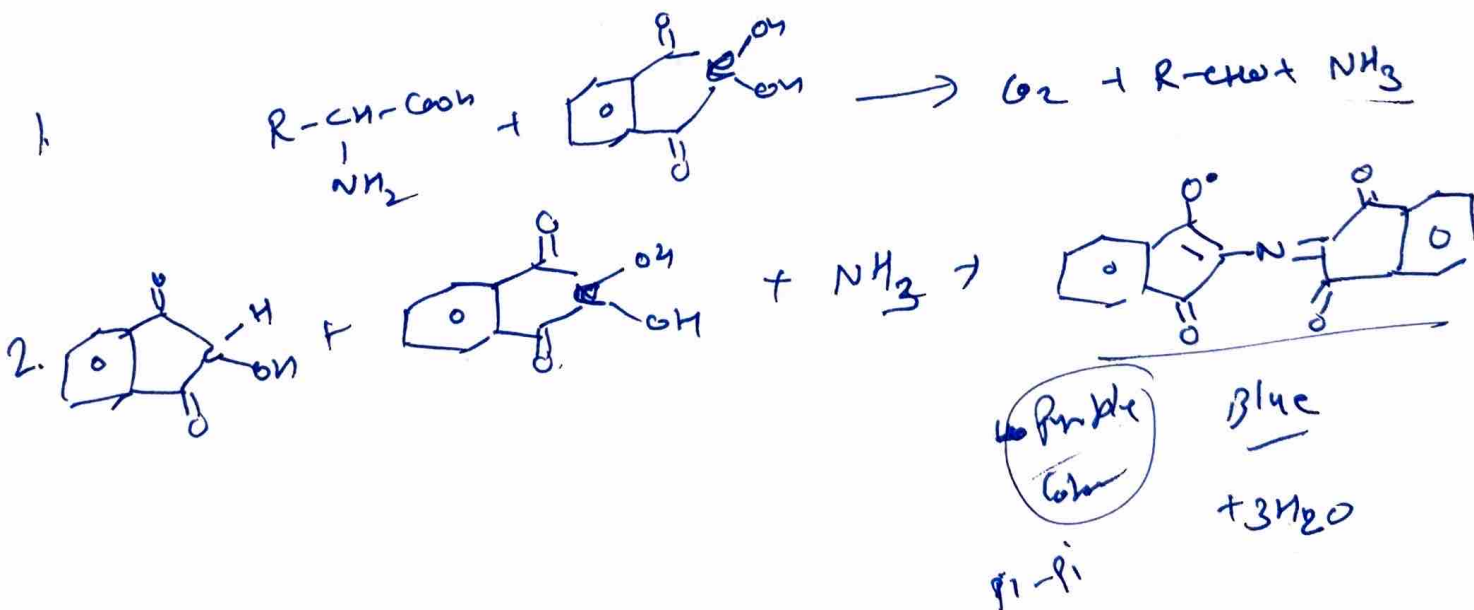
1. Terylene
2. Nylon
3. Cellulose
4. RNA

Column II

1. Glycosidic linkage
2. Ester linkage
3. Phosphodiester linkage
4. Amide linkage

1 → 2.
2 → 4
3 → 1
4 → 3

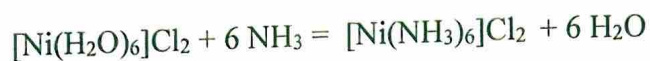
(viii) Amino acids are colorless compounds which show purple color on staining with ninhydrin. What is the purple color due to? Give its chemical structure with glycine as the amino acid. (2)



(Q3)

In one of your practical, you synthesized $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ from $\text{Ni}(\text{H}_2\text{O})_6\text{Cl}_2$.

- (i) Write the chemical equation for this reaction as performed in the lab. (1)



(Excess ammonia solution was taken, if you write 6 NH_3 or excess ammonia or NH_4OH or ammonia solution, marks will be given)

- (ii) Comment on the color of both complexes and explain the origin of their colors. (1)

$[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2 \rightarrow$ green (0.25 marks)

$[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2 \rightarrow$ blue (0.25 marks)

d-d transitions : (0.5 marks)

- (iii) Why do both $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ and $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ exhibit low-intensity color (low molar extinction coefficient)? (1)

Both have octahedral geometry and therefore the d-d transitions are Laporte forbidden.

- (iv) Which technique was used to purify $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$? (1)

Crystallization

- (v) What would be the consequence of using exactly six equivalents of NH_3 to convert $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ to $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$? (1)

As ammonia is a gas, it can escape from the reaction medium during the complex formation reaction. This may result in an incomplete reaction and the formation of a mixture of compounds, such as $[\text{Ni}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_2$ and $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$.

(Q4)

- (i) Write two methods for estimating acetic acid in a vinegar solution. (1)

- (1) Acid-base titration using a standard strong base such as NaOH / KOH
- (2) Using pH of the solution as K_a value of acetic acid is known

- (ii) What would be the color of phenolphthalein in CH_3COOH and NaOH solutions? (1)

(Q4)

- (i) Write two methods for estimating acetic acid in a vinegar solution. (1)

- (1) Acid-base titration using a standard strong base such as NaOH/ KOH
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- (ii) What would be the color of phenolphthalein in CH_3COOH and NaOH solutions? (1)

In a CH_3COOH solution phenolphthalein is colour less.

In NaOH solution phenolphthalein is pink to red coloured

- (iii) Write two important properties of an indicator used in acid-base titration. (1)

Any two points and the 1st one is must. (0.5 x 2)

Distinct Color Change: The indicator must exhibit a clear and sharp color change at the equivalence point to accurately signal the completion of the reaction.

pH Range Compatibility: Its pH transition range should closely match the pH range of the titration's equivalence point.

Stability: The indicator should remain stable and not degrade or react with the titrant, analyte, or other solution components.

Sensitivity: It must respond quickly and visibly to changes in pH, ensuring precise detection of the endpoint.

Non-Interference: The indicator should not interfere chemically with the reaction or alter the titration's outcome.

- (iv) pH of a vinegar solution was found to be 4.8. What is the concentration of acetic acid in vinegar solution? (1)
(Take pK_a of acetic acid as 4.8)

$$K_a = \frac{[H^+]^2}{[C]}$$

Rearranging to solve for $[C]$:

$$[C] = \frac{[H^+]^2}{K_a}$$

Substituting $[H^+] = 10^{-pH} = 10^{-4.8}$:

$$[H^+] = 1.58 \times 10^{-5}$$

$$[C] = \frac{(1.58 \times 10^{-5})^2}{1.58 \times 10^{-5}} = 1.58 \times 10^{-5} \text{ M}$$

Thus, the concentration of acetic acid in the solution is approximately ~~0.0158 M~~.

✓ $1.58 \times 10^{-5} \text{ M}$
 $15.8 \mu\text{M}$
 0.0158 mM ✓

- (v) 20 mL of 0.1 M acetic acid is mixed with 5 mL of 0.1 M NaOH. What will be pH of solution? (1)
 (Take pKa of acetic acid as 4.8)

$$pH = pK_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

The pH of the solution is approximately 4.32.